

Table 1. Flame Signal.

Flame Detector	Plug-in Amplifier	Honeywell Flame Safeguard Control	Minimum Acceptable Steady Current (microamps)	Maximum Current Expected (microamps)	Minimum Acceptable Voltage (Vdc)	Maximum Voltage (Vdc)
C7027A, C7035A, C7044A	R7249A	BC7000 + PM720	3.5	7.5	—	—
	R7749B (AMPLICHECK™)	BCS7700A	—	—	2.2	4.98
	R7249A	R4075C ^a , D ^a , E ^a	3.5	7.5	—	—
		R4138C ^a , D ^a	3.5	7.5	—	—
		R4140G, L, M	3.5	7.5	—	—
	R7290A	R4795A, D ^a	1.5	2.25	—	—
	None	R7023C ^a	1.5	2.25	—	—
	NONE	R7795A, C, E, G	1.5	2.25	—	—
	R7849A or R7849B (AMPLICHECK™)	7800 SERIES	—	—	1.25	5.0
	None	RA890G	1.5	2.25	—	—
C7927A	R7851B	7800 SERIES	—	—	1.25	5.0

^a Currently obsolete.

Ignition Spark Response Test

Test to be sure that ignition spark is not actuating the flame relay (usually 2K) or flame LED in the flame safeguard control.

1. Close the pilot and main burner manual fuel shutoff valves.
2. Connect the appropriate meter to the flame safeguard control amplifier. Start the burner and run through the ignition period. Ignition spark should occur, but the flame relay must not pull in or the flame LED should not light. The flame signal should not be more than 0.25 microamp or 1.25 Vdc with the BC7700 or 0.5 Vdc for the 7800 SERIES.
3. If the flame relay does pull in or the flame LED lights, reposition the flame detector to increase the distance between the flame detector and the ignition spark. If the flame detector is not in the line of sight of the ignition spark but appears to respond to the spark, it may be responding to reflected spark generated UV radiation. If so, relocate the flame detector so it does not receive the reflected UV radiation. It may be necessary to construct a barrier to block the UV radiation generated by the spark from the flame detector view.
4. Continue making the adjustments until the flame signal due to ignition spark is less than 0.25 microamp or 1.25 Vdc on the BC7700, or 0.5Vdc for the 7800 Series.
5. The use of the Q624 or Q652 solid-state ignition transformer may also provide a method to eliminate the C7027A, C7035A, or C7044A flame detector response to UV radiation generated by ignition spark. The Q624 and Q652 prevents flame detector response to ignition

spark by providing alternating periods of spark generation and UV sensor activation. If ignition spark is detected, try reversing the leads on the Q624 or Q652.

Response to Other Ultraviolet Sources

Some sources of artificial light produce small amounts of ultraviolet radiation. Under certain conditions, an ultraviolet detector will respond to them as if it is sensing a flame. Do not use an artificial light source to check the response of an ultraviolet detector. To check for proper flame detector operation, test for flame failure response under all operating conditions.

Weld The Sight Pipe (or Tighten the C7044A Clamp Screw)

When the flame signal is acceptable after all adjustments are made, remove the flame detector and weld the sight pipe in its final position. If you are using a swivel mount, the pipe may already be welded. Then reinstall the flame detector.

NOTE: If using a C7044A Detector with no sight pipe, do not remove the detector; tighten the clamp screw securely.

Final Checkout

Before putting the burner into service, check out the installation using the procedures in the Checkout section of the Instructions for the appropriate flame safeguard control. After completing the checkout, run the burner through at least one complete cycle to verify correct operation.

TROUBLESHOOTING

WARNING

Electrical Shock Hazard.

Can cause serious injury or death.

Be extremely careful while troubleshooting the detector; line voltage is present on some of the terminals when power is on.

Open the master switch to disconnect power before removing or installing the detector.

Inadequate Flame Signal

If a satisfactory flame signal cannot be obtained while adjusting the sighting position of the detector, perform the procedures indicated in steps 1 through 7. If other difficulties are encountered in the system, refer to Troubleshooting in the Instructions for the appropriate Honeywell flame safeguard control.

1. Check for proper line voltage. Make sure the master switch is closed, connections are correct, and power supply is the correct voltage and frequency.
2. Check the detector wiring for defects, including:
 - a. incorrect connections.
 - b. wiring type or size of wire.
 - c. deteriorated wire.
 - d. open circuits.
 - e. short circuits.
 - f. leakage paths caused by moisture, soot, or dirt.
3. With the burner running, check the temperature at the detector. When it exceeds 215°F (102°C) for a C7027, C7927 or C7044, or 250°F (121°C) for a C7035:
 - a. use a heat block (part no. 136733) if using a C7027A or C7927 Flame Detector.
 - b. add the additional insulation between the wall of the combustion chamber and the detector.
 - c. add a shield or screen to reflect radiated heat away from the detector, or
 - d. add cooling (refer to Sight Pipe Ventilation, page 6).
4. Remove the detector and clean the viewing window with a soft, clean cloth.

5. Clean the inside of the sight pipe (if one is used) before reinstalling the detector.
6. If the flame signal continues to be too low, replace the plug-in amplifier (if the control has one).
7. If you still cannot obtain a proper flame signal, replace flame detector (C7027A, C7927, C7044A) or UV radiation sensing tube (C7035A).

IMPORTANT:

At the completion of troubleshooting, be sure to perform the Adjustments and Checkout beginning on page 9.

MAINTENANCE

Periodic Maintenance

1. Clean the viewing window and sight pipe (if used) when necessary. Remove the detector and use a soft, clean cloth to remove accumulated contaminants from the UV radiation sensor tube glass envelope.
2. C7027, C7035 and C7044 ultraviolet sensing tubes have a life expectancy of 40,000 hours of continuous use within the ambient temperature and voltage ratings. Replace the sensing tube in the C7035, or replace the C7027, C7927 or C7044 Detector, at appropriate intervals.
3. Keep the flame detection system adjusted for the smoothest, most reliable operation as recommended by the burner manufacturer.

Cleaning the C7044A Detector

When necessary, clean the C7044 Flame Detector by using the following procedure:

1. Loosen the two screws holding the C7044 mounting bracket to the blast tube. To remove, twist the bracket and detector. Do not loosen the clamp screw that holds the mounting bracket to the detector.
2. Clean the viewing window with a soft, clean cloth.
3. Insert the detector into the mounting hole, and to realign it, twist against the mounting screws. Tighten the two mounting screws.

Automation and Control Solutions

Honeywell International Inc.

1985 Douglas Drive North

Golden Valley, MN 55422

Honeywell Limited-Honeywell Limitée

35 Dynamic Drive

Toronto, Ontario M1V 4Z9

customer.honeywell.com

Honeywell